Safety data sheet

BASF Safety data sheet according to Regulation (EC) No. 1907/2006 as amended from time to time.
Date / Revised: 16.12.2020 Version: 12.0
Date previous version: 22.06.2016 Previous version: 11.1
Product: **Sodium Nitrate HQ free flowing (non-food grade)**
(ID no. 30046439/SDS_GEN_GB/EN)
Date of print 17.12.2020

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Sodium Nitrate HQ free flowing (non-food grade)

UFI: YCJU-1013-Q004-HNQP

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Chemical
Recommended use: Raw material, process chemical, inorganic salts, Heat transfer agents, Agricultural industry, formulation agent

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF plc
4th and 5th Floors, 2 Stockport Exchange
Railway Road, Stockport, SK1 3GG
UNITED KINGDOM

Telephone: +44 161 485-6222
E-mail address: product-safety-north@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

Ox. Sol. 2	H272 May intensify fire; oxidizer.
Eye Dam./Irrit. 2	H319 Causes serious eye irritation.

According to BASF current knowledge and application of the criteria given in Annex I of Regulation (EC) No. 1272/2008, the following classification exceeding the classification given in Regulation (EC) No 1272/2008, Annex VI, Table 3.1 is required.

Ox. Sol. 2

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

Globally Harmonized System, EU (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H272	May intensify fire; oxidizer.
H319	Causes serious eye irritation.

Precautionary Statements (Prevention):

P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P280	Wear protective gloves and eye protection or face protection.
P220	Keep away from clothing and other combustible materials.

Precautionary Statements (Response):

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337 + P313	If eye irritation persists: Get medical attention.
P370 + P378	In case of fire: Use extinguishing powder, foam or CO2 for extinction.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to Regulation (EC) No 1272/2008 [CLP]

Hazard determining component(s) for labelling: Sodium nitrate

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

No specific dangers known, if the regulations/notes for storage and handling are considered.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

Sodium nitrate
$NaNO_3$

Contains: anticaking agent, free flowing agent, anticaking agent

Hazardous ingredients (GHS)
according to Regulation (EC) No. 1272/2008

Sodium nitrate

Content (W/W): >= 99 % CAS Number: 7631-99-4 EC-Number: 231-554-3 REACH registration number: 01-2119488221-41	Ox. Sol. 2 Eye Dam./Irrit. 2 H272, H319

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. After inhalation of decomposition products: Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Hazards: Danger of methaemoglobin formation after ingestion.

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), treat with toluonium chloride to reverse methaemoglobinanaemia. After inhalation of decomposition products: Pulmonary odema prophylaxis.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
water spray

Unsuitable extinguishing media for safety reasons:
ABC powder, carbon dioxide

5.2. Special hazards arising from the substance or mixture

Endangering substances: nitrogen oxides

Advice: The substances/groups of substances mentioned can be released if the product is involved in a fire.

5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Substance/product is an oxidizing agent and can supply oxygen to stimulate or accelerate the combustion of organic or other combustible substances/products.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

Use breathing apparatus if exposed to vapours/dust/aerosol.

6.2. Environmental precautions

Do not release untreated into natural waters.

6.3. Methods and material for containment and cleaning up

For large amounts: Sweep/shovel up. Dispose of absorbed material in accordance with regulations.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Keep container tightly sealed. Ensure suitable air extract/ventilation on process machinery and transportation equipment. Protect against moisture. Keep away from sources of ignition - No smoking.

Protection against fire and explosion:
The substance/product is non-combustible.

7.2. Conditions for safe storage, including any incompatibilities

Segregate from oxidizable substances. Segregate from reducing agents. Segregate from ammonium salts.

Suitable materials for containers: Stainless steel 1.4541, Stainless steel 1.4571, High density polyethylene (HDPE), Low density polyethylene (LDPE), Polyester resin, glass reinforced (Palatal A410), glass, enamelled, Carbon steel (Iron), rubberized, Aluminium
Further information on storage conditions: Keep container tightly closed.

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

No occupational exposure limits known.

PNEC
freshwater: 0.45 mg/l

marine water: 0.045 mg/l

intermittent release: 4.5 mg/l

STP: 18 mg/l

DNEL
worker:
Long-term exposure- systemic effects, Inhalation: 36.7 mg/m3

worker:
Long-term exposure- systemic effects, dermal: 20.8 mg/kg

consumer:
Long-term exposure- systemic effects, dermal: 12.5 mg/kg

consumer:
Long-term exposure- systemic effects, Inhalation: 10.9 mg/m3

consumer:
Long-term exposure- systemic effects, oral: 12.5 mg/kg

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed. Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1or FFP1)

Hand protection:
Chemical resistant protective gloves (EN 374)

Eye protection:
Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Handle in accordance with good industrial hygiene and safety practice. Do not breathe dust. Keep away from food, drink and animal feeding stuffs. No eating, drinking, smoking or tobacco use at the place of work. Take off immediately all contaminated clothing. Hands and/or face should be washed before breaks and at the end of the shift. Employees should shower at the end of the shift.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

Form:	crystalline, powder	
Colour:	white	
Odour:	odourless	
Odour threshold:		
	not applicable, odour not perceivable	
pH value:	8 - 9 (100 g/l, 20 °C)	
Melting point:	306 °C	
Boiling point:		
	(1,013.25 hPa) The substance / product decomposes therefore not determined.	
Flash point:		
	Study scientifically not justified.	
Evaporation rate:		
	The product is a non-volatile solid.	
Flammability:	not highly flammable	(other)
Lower explosion limit:		
	For solids not relevant for classification and labelling.	
Upper explosion limit:		
	For solids not relevant for classification and labelling.	
Vapour pressure:		
	The value has not be determined because of the high melting point.	
Density:	2.26 g/cm3 (20 °C) Literature data.	

Relative density: 2.26
Literature data.

Solubility in water:
874 g/l
(20 °C)

Partitioning coefficient n-octanol/water (log Kow):
Study scientifically not justified.

Thermal decomposition: > 600 °C
oxygen, Nitrogen, Disodium oxide

Viscosity, dynamic:
Study scientifically not justified.

Explosion hazard: not explosive

Fire promoting properties: Oxidizing. (Directive 92/69/EEC, A.17)

9.2. Other information

Self heating ability: It is not a substance capable of spontaneous heating.

Bulk density: approx. 1,300 kg/m3

pKA: 14.8
(25 °C)

Hygroscopy: hygroscopic

Adsorption/water - soil:
Study technically not feasible.

Surface tension:
Based on chemical structure, surface activity is not to be expected.

Molar mass: 84.99 g/mol

Angle of repose: 35 ° (trickle test (lab for material testing))

SECTION 10: Stability and Reactivity

10.1. Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No data available.

10.2. Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Peroxides: The product does not contain peroxides. The product/the substance has not a tendency towards the formation of peroxide.

10.3. Possibility of hazardous reactions
Reacts with reducing agents. Reacts with oxidizing agents.

10.4. Conditions to avoid

See SDS section 7 - Handling and storage. Avoid heating while in contact with easily oxidizable materials.

10.5. Incompatible materials

Substances to avoid:
reducing agents, oxidizable substances, ammonium compound

10.6. Hazardous decomposition products

Hazardous decomposition products:
Disodium oxide

SECTION 11: Toxicological Information

11.1. Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
There is a risk of damage to the blood (methemoglobinemia) after a single uptake of large quantities.

Experimental/calculated data:
LD50 rat (oral): 3,430 mg/kg (OECD Guideline 401)

(by inhalation):Study does not need to be conducted.

LD50 rat (dermal): > 5,000 mg/kg (OECD Guideline 402)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Irritation

Assessment of irritating effects:
Not irritating to the skin. Irritating to eyes.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Serious eye damage/irritation rabbit: no irreversible damage (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:

Mouse Local Lymph Node Assay (LLNA) mouse: Non-sensitizing. (OECD Guideline 429)

Germ cell mutagenicity

Assessment of mutagenicity:
The data available on mutagenic action are not consistent.

Carcinogenicity

Assessment of carcinogenicity:
In long-term studies in rats in which the substance was given by feed, a carcinogenic effect was not observed. Under certain conditions the substance can form nitrosamines. Nitrosamines are carcinogenic in animal studies.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The results were determined in a Screening test (OECD 421/422). The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies. The results were determined in a Screening test (OECD 421/422). The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Specific target organ toxicity (single exposure)

No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The substance may cause damage to the hematological system after repeated ingestion.

Aspiration hazard

Study does not need to be conducted.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:
LC50 (96 h) 7,950 mg/l, Oncorhynchus tschawytscha (static)
Literature data. Nominal concentration.

Aquatic invertebrates:
EC50 (24 h) 8,609 mg/l, Daphnia magna (Daphnia test acute, static)

Aquatic plants:
EC50 (10 d) > 1,700 mg/l (chlorophyll content), algae (static)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Microorganisms/Effect on activated sludge:
EC10 (3 h) 180 mg/l, activated sludge, domestic (OECD Guideline 209, aquatic)

Chronic toxicity to fish:
Study scientifically not justified.

Chronic toxicity to aquatic invertebrates:
Study scientifically not justified.

Assessment of terrestrial toxicity:
No data available.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Not applicable for inorganic substances. Can be oxidized to nitrate, or be reduced to nitrogen, by microorganisms.

Elimination information:
not applicable

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.
Study scientifically not justified.

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Accumulation in organisms is not to be expected.

Bioaccumulation potential:
Study scientifically not justified.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): PBT assessment does not apply. Not applicable for inorganic substances.

12.6. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.7. Additional information

Other ecotoxicological advice:
Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Contact manufacturer regarding recycling.
Check for possible recycling.
Contact waste centre regarding recycling.
Test for use in agriculture.

The UK Environmental Protection (Duty of Care) Regulations (EP) and amendments should be noted (United Kingdom).
This product and any uncleaned containers must be disposed of as hazardous waste in accordance with the 2005 Hazardous Waste Regulations and amendments (United Kingdom)

SECTION 14: Transport Information

Land transport

ADR

UN number	UN1498
UN proper shipping name:	SODIUM NITRATE
Transport hazard class(es):	5.1
Packing group:	III
Environmental hazards:	no

Special precautions for user:	Tunnel code: E

RID

UN number	UN1498
UN proper shipping name:	SODIUM NITRATE
Transport hazard class(es):	5.1
Packing group:	III
Environmental hazards:	no
Special precautions for user:	None known

<u>Inland waterway transport</u>
ADN

UN number	UN1498
UN proper shipping name:	SODIUM NITRATE
Transport hazard class(es):	5.1
Packing group:	III
Environmental hazards:	no
Special precautions for user:	None known

<u>Transport in inland waterway vessel</u>
Not evaluated

<u>Sea transport</u>

IMDG

UN number:	UN 1498
UN proper shipping name:	SODIUM NITRATE
Transport hazard class(es):	5.1
Packing group:	III
Environmental hazards:	no Marine pollutant: NO
Special precautions for user:	None known

<u>Air transport</u>

IATA/ICAO

UN number:	UN 1498
UN proper shipping name:	SODIUM NITRATE

Transport hazard class(es):	5.1
Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

14.1. UN number
See corresponding entries for “UN number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Transport in bulk according to Annex II of MARPOL and the IBC Code

Regulation:	Not evaluated
Shipment approved:	Not evaluated
Pollution name:	Not evaluated
Pollution category:	Not evaluated
Ship Type:	Not evaluated

Further information
This product is subject to the most recent edition of "The Carriage of Dangerous Goods and Use of Transportable Pressure Equipment Regulations" and their amendments (United Kingdom).

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

The data should be considered when making any assessment under the Control of Substances Hazardous to Health Regulations (COSHH), and related guidance, for example, 'COSHH Essentials' (United Kingdom).
This product may be subject to the Control of Major Accident Hazards Regulations (COMAH), and amendments if specific threshold tonnages are exceeded (United Kingdom).

15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

SECTION 16: Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. Any other intended applications should be discussed with the manufacturer.

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

Ox. Sol.	Oxidising solids
Eye Dam./Irrit.	Serious eye damage/eye irritation
H272	May intensify fire; oxidizer.
H319	Causes serious eye irritation.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.

Annex: Exposure Scenarios

Index

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1. Short title of exposure scenario

Industrial applications, Manufacture of substance
SU3; SU8; ERC1; PROC1, PROC2, PROC3, PROC4, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC1: Use in closed process, no likelihood of exposure. PROC2: Use in closed, continuous process with occasional controlled exposure. PROC3: Use in closed batch process (synthesis or formulation). PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. PROC8a: Transfer of substance or preparation (charging/discharging) from/to ves-sels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC15: Use a laboratory reagent. Use domain: industrial
Operational conditions	
Concentration of the substance	Sodium nitrate Content: >= 0 % - <= 100 %

Physical state	Solid
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Ensure segregation of worker from the source Ensure minimization of manual phases Avoid skin contact. Avoid contact with eyes. Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Minimise number of staff exposed.	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Ensure containment of the emission source	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
Segregate substance from incompatible materials.	

Contributing exposure scenario	
Use descriptors covered	As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.

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2. Short title of exposure scenario

Industrial applications, Distribution of substance, (use in industrial settings)
SU3; SU3, SU10; ERC2, ERC4, ERC5, ERC6a, ERC6b, ERC7; PROC1, PROC2, PROC3, PROC4, PROC5, PROC7, PROC8a, PROC8b, PROC9, PROC10, PROC12, PROC13, PROC14, PROC15, PROC19, PROC20, PROC22, PROC23, PROC24, PROC26; PC1, PC4, PC11, PC12, PC14, PC16, PC17, PC19, PC20, PC35, PC37, PC0, PC10

Control of exposure and risk management measures

Contributing exposure scenario

Use descriptors covered	PROC1: Use in closed process, no likelihood of exposure. PROC2: Use in closed, continuous process with occasional controlled exposure. PROC3: Use in closed batch process (synthesis or formulation). PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. PROC8a: Transfer of substance or preparation (charging/discharging) from/to ves-sels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC15: Use a laboratory reagent. Use domain: industrial
Operational conditions	
Concentration of the substance	Sodium nitrate Content: >= 0 % - <= 100 %
Physical state	Solid
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Ensure segregation of worker from the source Ensure minimization of manual phases Avoid skin contact. Avoid contact with eyes. Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Minimise number of staff exposed.	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Ensure containment of the emission source	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
Segregate substance from incompatible materials.	

Contributing exposure scenario	
Use descriptors covered	PROC1: Use in closed process, no likelihood of exposure. PROC2: Use in closed, continuous process with occasional controlled exposure. PROC3: Use in closed batch process (synthesis or formulation). PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises. PROC8a: Transfer of substance or preparation (charging/discharging) from/to ves-sels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC15: Use a laboratory reagent. Use domain: industrial
Operational conditions	
Concentration of the substance	Sodium nitrate Content: >= 0 % - <= 100 %
Physical state	liquid
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Ensure segregation of worker from the source Ensure minimization of manual phases Avoid skin contact. Avoid contact with eyes. Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Minimise number of staff exposed.	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Ensure containment of the emission source	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
Segregate substance from incompatible materials.	

Contributing exposure scenario	
Use descriptors covered	PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). PROC7: Industrial spraying PROC10: Roller application or brushing PROC12: Use of blow agents in manufacture of foam. PROC13: Treatment of articles by dipping and pouring. PROC14: Production of preparations or articles by tabletting, compression, extrusion, pelettisation. PROC19: Hand-mixing with intimate contact and only PPE available. PROC20: Heat and pressure transfer fluids in dispersive use but closed systems PROC22: Potentially closed processing operations (with minerals) at elevated temperature PROC23: Open processing and transfer operations (with minerals) at elevated temperature PROC24: High (mechanical) energy work-up of substances bound in materials and/or articles PROC26: Handling of solid inorganic substances at ambient temperature. Use domain: industrial
Operational conditions	
Concentration of the substance	Sodium nitrate Content: >= 0 % - <= 100 %
Physical state	Solid
Physical state	Liquid, low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Risk Management Measures	
Ensure segregation of worker from the source Ensure minimization of manual phases Avoid skin contact. Avoid contact with eyes. Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Minimise number of staff exposed.	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Ensure containment of the emission source	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk	

characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
Segregate substance from incompatible materials.	

Contributing exposure scenario	
Use descriptors covered	As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.

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3. Short title of exposure scenario

Professional applications
SU22; SU22; ERC8a, ERC8b, ERC8c, ERC8d, ERC8e, ERC8f, ERC9a, ERC9b; PROC2, PROC3, PROC5, PROC8a, PROC8b, PROC9, PROC10, PROC11, PROC13, PROC19, PROC20, PROC26; PC1, PC4, PC11, PC12, PC14, PC16, PC17, PC20, PC37, PC0, PC10

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	PROC2: Use in closed, continuous process with occasional controlled exposure. PROC3: Use in closed batch process (synthesis or formulation). PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/or significant contact). PROC8a: Transfer of substance or preparation (charging/discharging) from/to ves-sels/large containers at non-dedicated facilities PROC8b: Transfer of substance or preparation (charging/discharging) from/to vessels/large containers at dedicated facilities PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). PROC10: Roller application or brushing PROC11: Non industrial spraying PROC13: Treatment of articles by dipping and pouring. PROC19: Hand-mixing with intimate contact and only PPE available. PROC20: Heat and pressure transfer fluids in dispersive use but closed systems PROC26: Handling of solid inorganic substances at ambient temperature. Use domain: professional
Operational conditions	
Concentration of the substance	Sodium nitrate Content: >= 0 % - <= 100 %
Physical state	Solid

Physical state	Liquid, low fugacity
Duration and Frequency of activity	Exposure duration: 480 min 5 days per week
Indoor/Outdoor	Indoor
Indoor/Outdoor	Outdoor
Risk Management Measures	
Avoid splashing. Ensure that no inhalable dusts are generated. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid skin contact. Avoid contact with eyes. Clean equipment and the work area every day. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Minimise number of staff exposed.	
Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Ensure containment of the emission source	
Use suitable eye protection.	
Risk Management Measures are based on qualitative risk characterisation.	
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Worker - contact with eyes
Additional good practice advice	
Segregate substance from incompatible materials.	

Contributing exposure scenario	
Use descriptors covered	As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.

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4. Short title of exposure scenario

Consumer applications
SU21; SU21; ERC8a, ERC8b, ERC8c, ERC8d, ERC8e, ERC8f, ERC9a, ERC9b, ERC10a, ERC11a; PC1, PC4, PC12, PC16, PC17, PC35, PC39, PC0, PC10

Control of exposure and risk management measures

Contributing exposure scenario

Use descriptors covered	SU21: Consumer uses All relevant product categories
Operational conditions	
Physical state	liquid
Risk Management Measures	
Consumer Measures	Use suitable eye protection.
Exposure estimate and reference to its source	
Assessment method	Qualitative assessment
	Consumer - contact with eyes
	Contact is only accidental.

Contributing exposure scenario	
Use descriptors covered	As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed.

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